

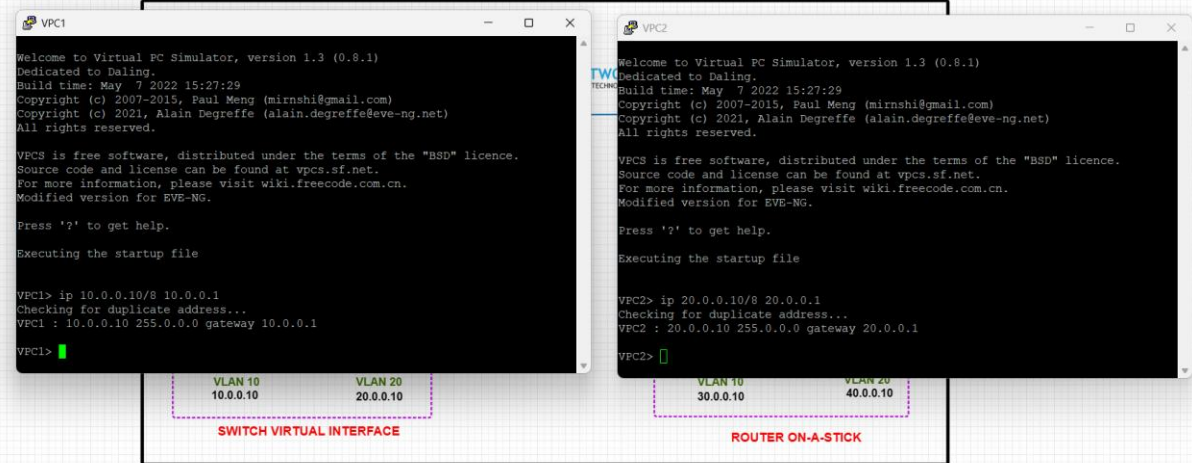
LAB 4 INTER VLAN

GURUCHANDHRAN M

TASK 1: PC IP ASSIGNMENT

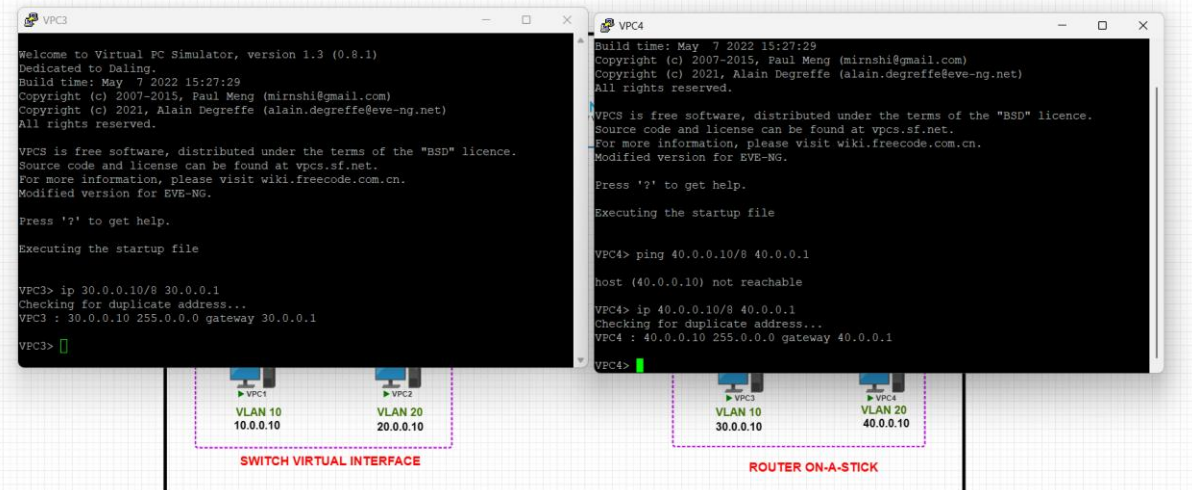
PC1 IP: 10.0.0.10

PC2 IP: 20.0.0.10

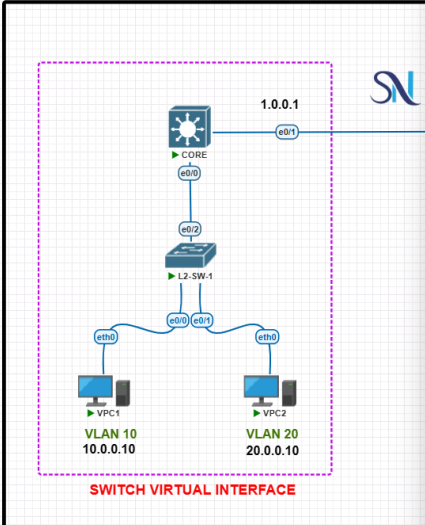


PC3 IP: 30.0.0.10

PC4 IP: 40.0.0.10



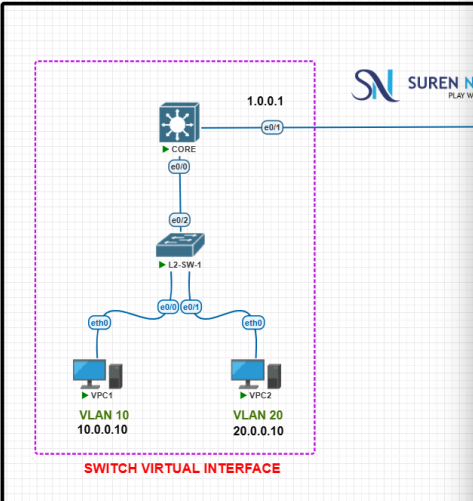
TASK 2: LAYER 2 SWITCH CONFIGURATION IN SVI



The network diagram shows a topology with a CORE switch at the top, connected to L2-SW-1 below it. L2-SW-1 has two Ethernet interfaces (e0/0 and e0/1) connected to two Virtual PCs (VPC1 and VPC2). VPC1 is in VLAN 10 with IP 10.0.0.10, and VPC2 is in VLAN 20 with IP 20.0.0.10. The CORE switch has an interface e0/1 connected to L2-SW-1's e0/0. The CORE switch has an IP of 1.0.0.1. The L2-SW-1 switch is highlighted with a dashed purple box and labeled 'SWITCH VIRTUAL INTERFACE' in red. A watermark 'SN SUREN' is visible in the background.

```
L2-SW-1#
L2-SW-1#CONF T
Enter configuration commands, one per line. End with CNTL/Z.
L2-SW-1(config)#vlan 10
L2-SW-1(config-vlan)#ex
L2-SW-1(config)#vlan 20
L2-SW-1(config-vlan)#ex
L2-SW-1(config)#int e0/0
L2-SW-1(config-if)#switchport vlan 10
^
% Invalid input detected at '^' marker.
L2-SW-1(config-if)#switchport access vlan 10
L2-SW-1(config-if)#ex
L2-SW-1(config)#int e0/1
L2-SW-1(config-if)#switchport access vlan 20
L2-SW-1(config-if)#ex
L2-SW-1(config)#int e0/2
L2-SW-1(config-if)#switchport access encapsulation dot1q
^
% Invalid input detected at '^' marker.
L2-SW-1(config-if)#switchport access trunk encapsulation dot1q
^
% Invalid input detected at '^' marker.
L2-SW-1(config-if)#switchport trunk encapsulation dot1q
L2-SW-1(config-if)#switchport mode trunk
L2-SW-1(config-if)#ex
L2-SW-1(config)#
*Aug 14 09:44:30.785: %FNP-6-FNP_DISCOVERY_STOPPED: PnP Discovery stopped (Config Wizard)
L2-SW-1(config)#
```

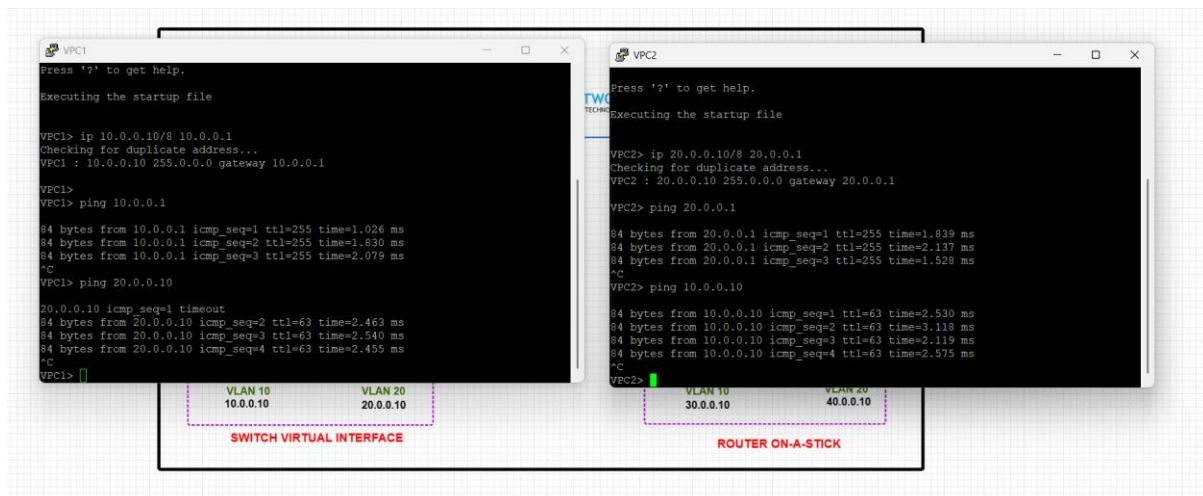
TASK 3: LAYER 3 - CORE SWITCH CONFIGURATION IN SVI



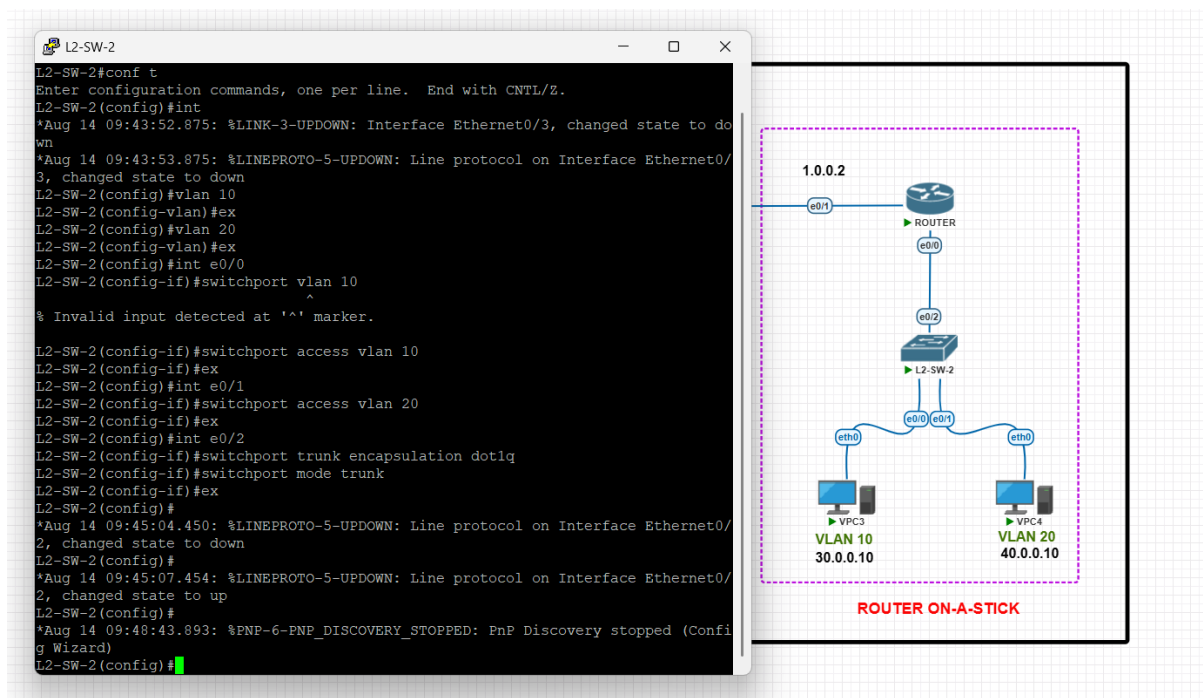
The network diagram is identical to the one in Task 2, showing the CORE switch, L2-SW-1, and two VPCs. The CORE switch is highlighted with a dashed purple box and labeled 'SWITCH VIRTUAL INTERFACE' in red. A watermark 'SN SUREN' is visible in the background.

```
CORE#
CORE(config)#int vlan 10
CORE(config-if)#ip add
*Aug 14 09:47:09.524: %LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan10, changed state to down
CORE(config-if)#ip add 10.0.0.1 255.0.0.0
CORE(config-if)#no sh
CORE(config-if)#ex
CORE(config)#int vlan 20
*Aug 14 09:47:21.970: %LINK-3-UPDOWN: Interface Vlan10, changed state to up
*Aug 14 09:47:22.970: %LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan10, changed state to up
CORE(config)#int vlan 20
*Aug 14 09:47:25.152: %LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan20, changed state to down
CORE(config-if)#ip add 20.0.0.1 255.0.0.0
CORE(config-if)#no sh
CORE(config-if)#ex
CORE(config)#
*Aug 14 09:47:42.399: %LINK-3-UPDOWN: Interface Vlan20, changed state to up
*Aug 14 09:47:43.399: %LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan20, changed state to up
CORE(config)#ip Routing
CORE(config)#int e0/1
CORE(config-if)#ip add 1.0.0.1 255.0.0.0
^
% Invalid input detected at '^' marker.
CORE(config-if)#ip 1.0.0.1 255.0.0.0
^
% Invalid input detected at '^' marker.
CORE(config-if)#no switchport
CORE(config-if)#ip 1.0.0.1 255.0.0.0
^
% Invalid input detected at '^' marker.
CORE(config-if)#ip 1.0.0.1 255.0.0.0
*Aug 14 09:48:30.260: %LINK-3-UPDOWN: Interface Ethernet0/1, changed state to up
*Aug 14 09:48:31.271: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet0/1, changed state to up
CORE(config-if)#ip add 1.0.0.1 255.0.0.0
CORE(config-if)#no sh
CORE(config-if)#ex
CORE(config)#
```

TASK 4: CONNECTION ESTABLISHMENT VERIFICATION IN SVI USING ping



TASK 5: LAYER 2 SWITCH CONFIGURATION IN ROUTER ON A STACK



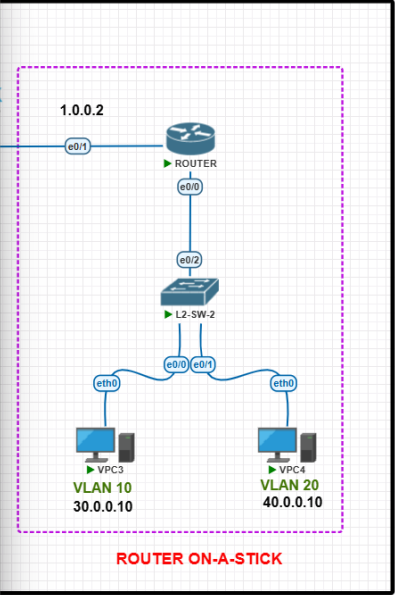
TASK 6: ROUTER CONFIGURATION IN ROUTER ON A STACK

```
ROUTER
ROUTER(config)#int e0/0
ROUTER(config-if)#no sh
ROUTER(config-if)#ex
ROUTER(config)#int e0/0.1
ROUTER(config-subif)#ip add 30.0.0.1 255.0.0.0

% Configuring IP routing on a LAN subinterface is only allowed if that
subinterface is already configured as part of an IEEE 802.1Q, IEEE 802.1Q,
or ISL VLAN.

ROUTER(config-subif)#ex
ROUTER(config)#int e0/0.1
ROUTER(config-subif)#encapsulation dot1q 10
ROUTER(config-subif)#ip add 30.0.0.1 255.0.0.0
ROUTER(config-subif)#no sh
ROUTER(config-subif)#ex
ROUTER(config)#int e0/0.2
ROUTER(config-subif)#encapsulation dot1q 20
*Aug 14 10:31:33.056: %PNP-6-PNP_DISCOVERY_STOPPED: PnP Discovery stopped (Config Wizard)
ROUTER(config-subif)#ip add 40.0.0.1 255.0.0.0
ROUTER(config-subif)#no sh
ROUTER(config-subif)#ex
ROUTER(config)#
ROUTER(config)#int e0/1
ROUTER(config-if)#ip add 1.0.0.2 255.0.0.0
ROUTER(config-if)#no sh
ROUTER(config-if)#ex
ROUTER(config)#do show ip int br
Interface              IP-Address      OK? Method Status  Prot
ocol
Ethernet0/0            unassigned      YES TFTP  up      up
Ethernet0/0.1          30.0.0.1        YES manual up      up
Ethernet0/0.2          40.0.0.1        YES manual up      up
Ethernet0/1            1.0.0.2         YES manual up      up
Ethernet0/2            unassigned      YES TFTP  down    down
Ethernet0/3            unassigned      YES TFTP  down    down

ROUTER(config)#
```



Physical

TASK 7: CONNECTION ESTABLISHMENT VERIFICATION IN ROS USING ping

```
VPC3> ip 30.0.0.10/8 30.0.0.1
Checking for duplicate address...
VPC3 : 30.0.0.10 255.0.0.0 gateway 30.0.0.1

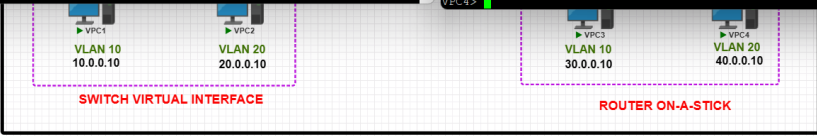
VPC3>
VPC3> ping 30.0.0.1
host (30.0.0.1) not reachable

VPC3> ping 30.0.0.1
84 bytes from 30.0.0.1 icmp_seq=1 ttl=255 time=0.716 ms
84 bytes from 30.0.0.1 icmp_seq=2 ttl=255 time=1.377 ms
84 bytes from 30.0.0.1 icmp_seq=3 ttl=255 time=1.206 ms
^C
VPC3> ping 40.0.0.10
84 bytes from 40.0.0.10 icmp_seq=1 ttl=63 time=3.092 ms
84 bytes from 40.0.0.10 icmp_seq=2 ttl=63 time=3.094 ms
84 bytes from 40.0.0.10 icmp_seq=3 ttl=63 time=2.570 ms
^C
VPC3>
```

```
VPC4> host (40.0.0.10) not reachable
VPC4> ip 40.0.0.10/8 40.0.0.1
Checking for duplicate address...
VPC4 : 40.0.0.10 255.0.0.0 gateway 40.0.0.1

VPC4>
VPC4> ping 40.0.0.1
host (40.0.0.1) not reachable

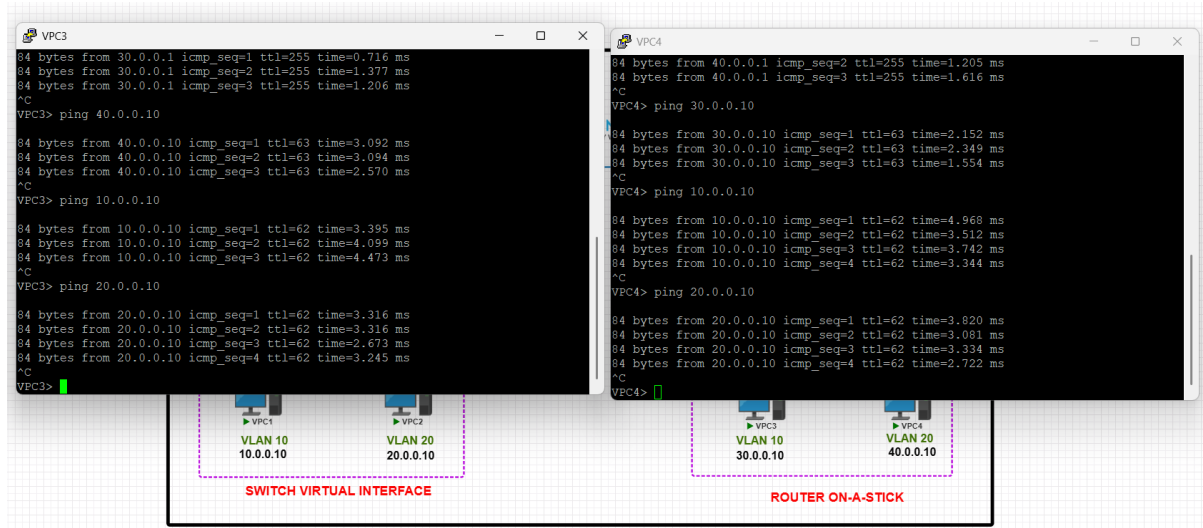
VPC4> ping 40.0.0.1
84 bytes from 40.0.0.1 icmp_seq=1 ttl=255 time=1.774 ms
84 bytes from 40.0.0.1 icmp_seq=2 ttl=255 time=1.205 ms
84 bytes from 40.0.0.1 icmp_seq=3 ttl=255 time=1.616 ms
^C
VPC4> ping 30.0.0.10
84 bytes from 30.0.0.10 icmp_seq=1 ttl=63 time=2.152 ms
84 bytes from 30.0.0.10 icmp_seq=2 ttl=63 time=2.349 ms
84 bytes from 30.0.0.10 icmp_seq=3 ttl=63 time=1.554 ms
^C
VPC4>
```



ROUTER ON-A-STICK

TASK 7: CONNECTION ESTABLISHMENT VERIFICATION FROM SVI TO ROS

From ROS to SVI



From SVI to ROS

